

BPL_TEST2_Chemostat - demo

```
In [1]: # Import the application model and context variables
        from BPL_TEST2_Chemostat import*
```

Windows - run FMU pre-compiled JModelica 2.14

```
In [2]: # Import the general FMU_explore functions and adapt them to the application
        from fmu_explore_pyfmi import fmu_explore

        Dummy = fmu_explore(model, parValue, parLocation, parCheck, fmu_model, fmu_proc,
                             MSL_usage, MSL_version, BPL_version, \
                             opts_std, simulationTime, timeDiscreteStates, stateValue, \
                             diagrams, ax, lines, external_function = cstrProdMax)

        par = Dummy.par
        init = Dummy.init
        disp = Dummy.disp
        simu = Dummy.simu
        show = Dummy.show
        resetPen = Dummy.resetPen
        process_diagram = Dummy.process_diagram
        describe_general = Dummy.describe_general
        describe_parts = Dummy.describe_parts
        describe_MSL = Dummy.describe_MSL
        FMU_explore_info = Dummy.FMU_explore_info
        system_info = Dummy.system_info
        SDG = Dummy.SDG
        cstrProdMax = Dummy.external_function
```

Bring in the user defined FMU_explore functions describe() and newplot()

```
In [3]: run -i BPL_TEST2_Chemostat_explore.py
```

Model for the process has been setup. Key commands:

- par() - change of parameters and initial values
- init() - change initial values only
- simu() - simulate and plot
- newplot() - make a new plot
- show() - show plot from previous simulation
- disp() - display parameters and initial values from the last simulation
- describe() - describe culture, broth, parameters, variables with values/units

Note that both disp() and describe() takes values from the last simulation and the command process_diagram() brings up the main configuration

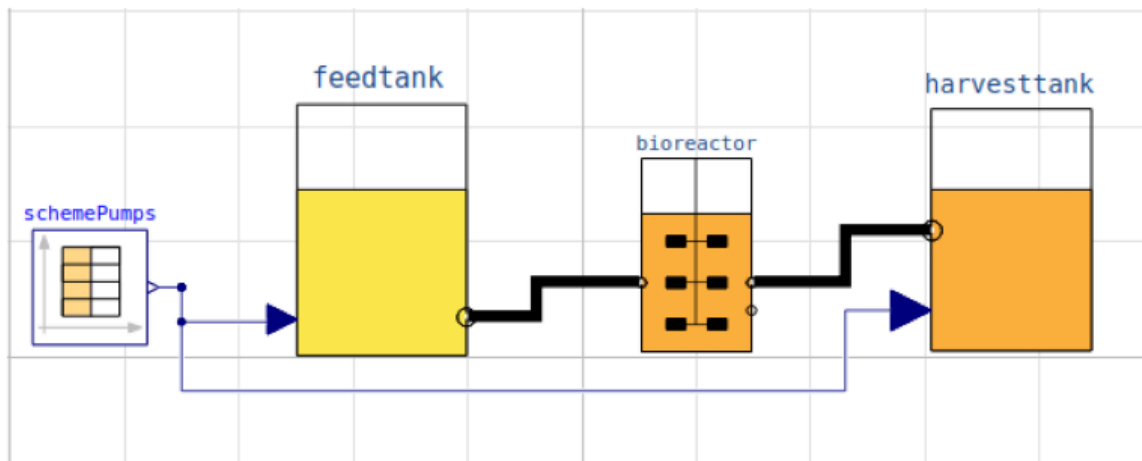
Brief information about a command by help(), eg help(simu)

Key system information is listed with the command system_info()

```
In [4]: %matplotlib inline
        plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]
```

```
In [5]: process_diagram()
```

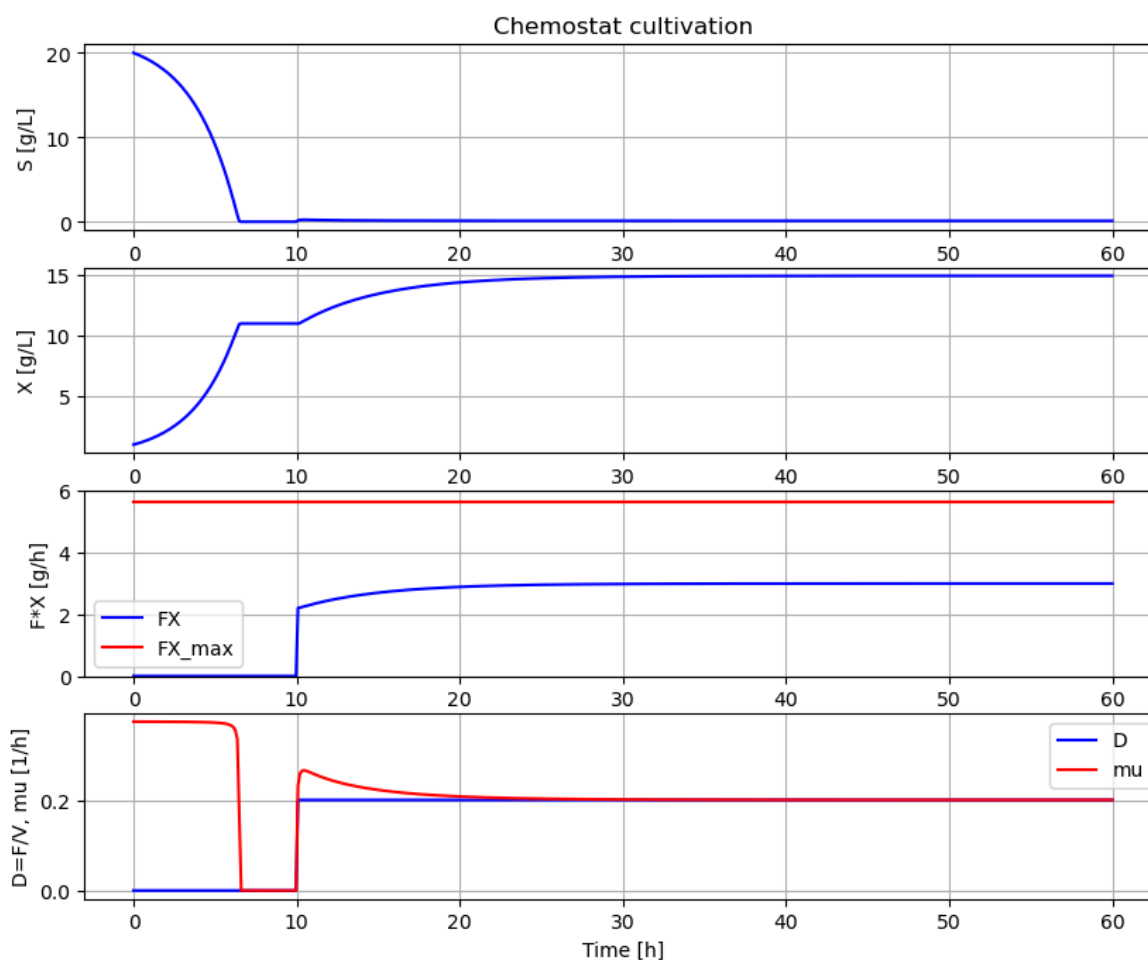
No processDiagram.png file in the FMU, but try the file on disk.



```
In [6]: #fmu_model = 'xBPL_TEST2_Chemostat_linux_om_me.fmu'
#model = load_fmu(fmu_model, log_level=0)
```

```
In [7]: newplot(); ax[2].set_ylim([0, 6])
par(Y=0.50, qSmax=0.75, Ks=0.1)           # Culture parameters
init(V_start=1.0, VX_start=1.0, VS_start=20) # Bioreactor startup
par(S_in=30, t0=0, F0=0, t1=10, F1=0.2)    # Substrate feeding
simu(60)
```

```
Out[7]: <pyfmi.fmi_algorithm_drivers.FMIResult at 0x16d8595bd40>
```



```
In [8]: # The maximal biomass productivity FX_max [g/h] marked red in the diagram above
# can be calculated for CSTR from the FMU and is
cstrProdMax(model)
```

```
Out[8]: np.float64(5.625)
```

```
In [9]: describe('cstrProdMax')
```

Calculate from the model maximal chemostat productivity FX_max : 5.625 [g/h]

```
In [10]: disp('culture')
```

Y : 0.5
qSmax : 0.75
Ks : 0.1

```
In [11]: describe('mu')
```

Cell specific growth rate variable : 0.2 [1/h]

```
In [12]: describe('parts')
```

['bioreactor', 'bioreactor.culture', 'D', 'feedtank', 'harvesttank', 'MSL', 'schemePumps']

```
In [13]: describe('MSL')
```

MSL: RealInput, RealOutput, CombiTimeTable, Types

```
In [14]: system_info()
```

System information
-OS: Windows
-Python: 3.12.11
-Scipy: not installed in the notebook
-PyFMI: 2.21.0
-FMU by: JModelica.org
-FMI: 2.0
-Type: FMUModelCS2
-Name: BPL.Examples_TEST2.Chemostat
-Generated: 2025-07-26T09:39:44
-MSL: 3.2.2 build 3
-Description: Bioprocess Library version 2.3.1
-Interaction: FMU-explore version 1.1.4

```
In [15]: cstrProdMax()
```

```
Out[15]: np.float64(5.625)
```

```
In [ ]:
```